



The Iowa Water Problem:

Predicting Nitrate Water Violations in Iowa
using XGBoost

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Erdős Institute, Summer 2026

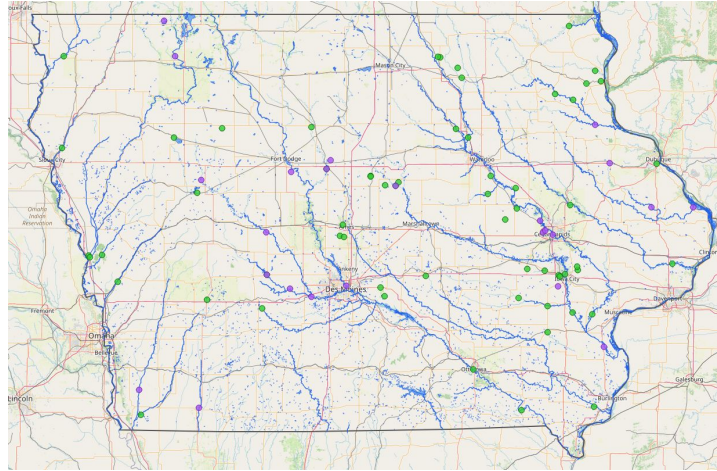


The Problem

- Iowa grows 2.5 billion bushels of corn each year
 - Enough calories to feed the entire US
- Cancer rates in Iowa are **rising**
 - Iowa is the only state in the US where cancer rates are rising, and this trend has held since 2015
- The link: nitrogen fertilizer → nitrate runoff → drinking water
 - Nitrate is a Group 2A carcinogen
 - Federal limit: 10 mg/L. Background: 1–2 mg/L. Elevated risk starts around 3 mg/L.

Why this is hard

- Real-time sensors exist — but coverage is sparse, expensive, and shrinking (budget cuts)
- This leaves stakeholders (**water utilities, regulators, farmers**) flying blind

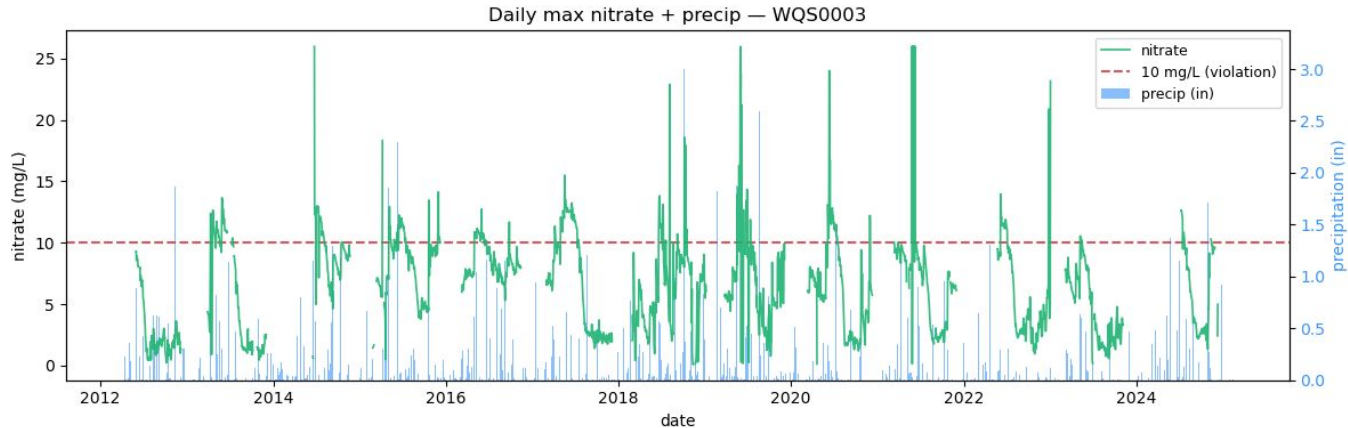


Goal: flag risk at any location

Research Question: Can we predict nitrate risk at a specific location using ONLY publicly available weather and land-use data, without a physical sensor?

Data story: the sensors

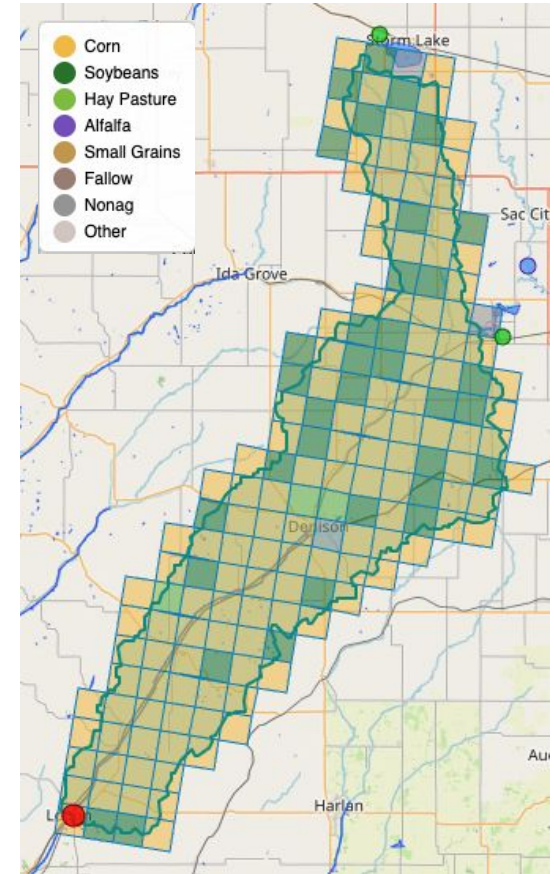
- 162 candidate sensors → 85 kept after filtering (2008–2025, 4–15 yrs of records each)



Data story: the basins

- Drainage basin computed per sensor
- Layered on:
 - Daily weather data: Precipitation (inches), Temperature (°C), Solar Radiation (W/m²), and more
 - Annual Crop Data (30m resolution from satellites)
 - Soil Nitrogen Surplus Data (250m resolution, computed in gTREND-Nitrogen Nature paper)

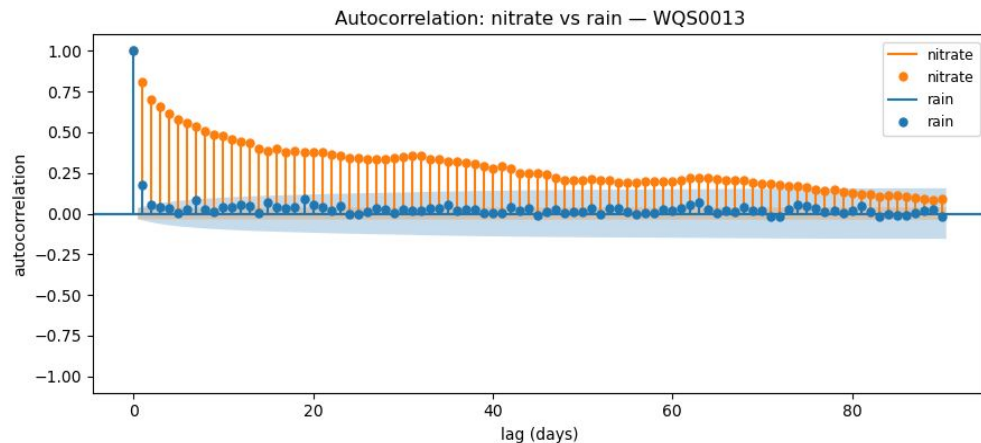
Drainage basin of WQS0040



EDA Highlights

Nitrate has a strong memory: it's highly autocorrelated. This means that if you have access to a site's history, “predict yesterday” is a high unbeatable baseline. ($R^2 \approx 0.92$ for regression)

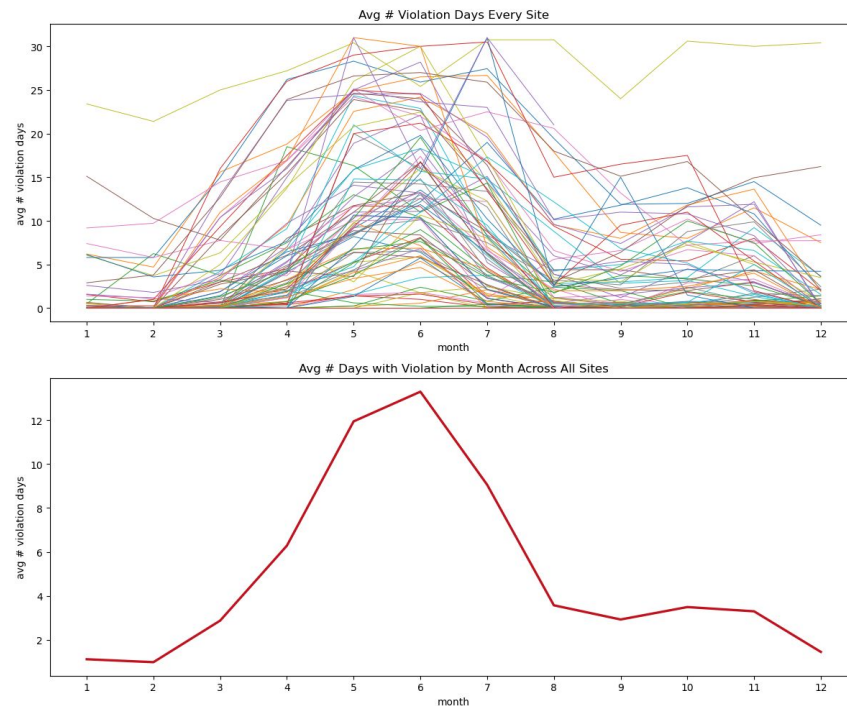
...but virtual sites don't have site history – there's no sensor there.



EDA Highlights

Nitrate is highly seasonal: it starts to climb in April and falls later in the summer. This seasonality holds even after controlling for the seasonality of weather.

We had to ensure our model learned more than “predict high nitrate in June”.

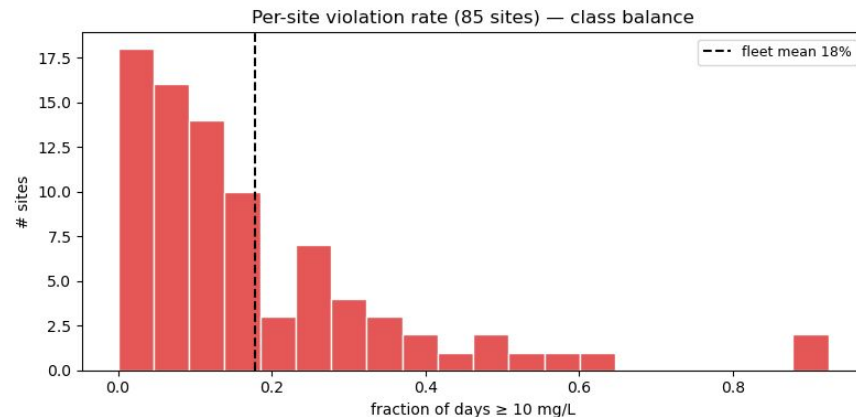
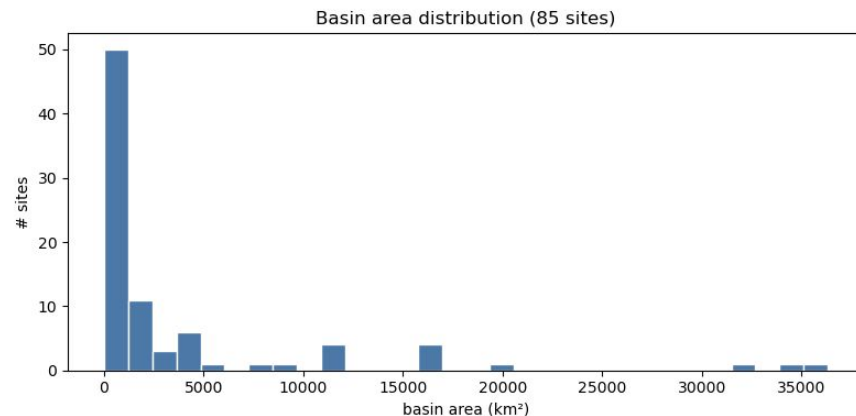


Average # of violation days by month

EDA Highlights

Site variability is extreme in both geography and nitrate severity.

- Most basins are relatively small, but there are several sensors placed in major rivers with mega basins.
- Some sites frequently saw nitrate levels as high as 35 mg/L, others almost never witnessed a violation.



Distribution of basin size (by area) and violation rate



Feature Engineering

- **Land far from the sensor matters less than land near the sensor.** Aggregated data into “distance buckets” and further weighted values by their specific to-sensor distance.
- Static features matter a lot. Latitude and longitude seem to help model
- Kept: crop mix near the sensor, recent (not same-day) rainfall, sensor lat/lon, rest-of-state nitrate (leakage-safe), day-of-year sin/cos
- Created: distance "buckets" instead of raw basin averages — 2 buckets beat exponential decay and 3 buckets
- Dropped: rolling precipitation/water-balance features (hurt performance), broadcasting stale 2017 nitrogen-surplus values forward (hurt performance)



Final Feature List:

Land far from the sensor matters less than land near the sensor. Aggregated data into “distance buckets” and further weighted values by their specific to-sensor distance.

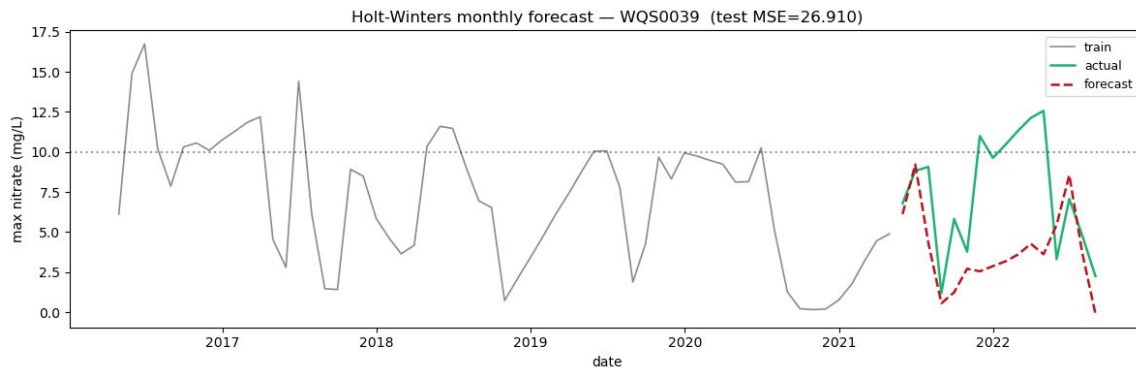
- Weather data (precipitation, temperature, solar radiation, etc) lagged by distance to sensor
- Annual crop data (corn, soybeans, alfalfa, hay pasture, small grains, fallow, non-ag and other)
- Annual nitrogen soil surplus (kg/acre)
- Geographical features: lat/lon, basin area, mean distance to sensor
- Calendar day information
- Cross-site nitrate: average nitrate at all sites except this one
- 68 feature columns total

Classifier models got 3 distance buckets (0-2 km, 2-5 km, 5+ km) and regression models got 2 distance buckets (0-5 km, 5+ km) as per our tuning.

Modeling Journey: Traditional Time Series

Traditional time series forecasting doesn't make sense for our goal, since virtual sites by definition don't have past values to use in forecasting.

Here's a graph showing a Holt-Winters exponential smoothing time series model for one site, but we quickly moved on.





Modeling Journey: Single site isolation

Modeling individual sites in isolation doesn't work, nor is it useful as stated.

- Even gradient boosting and random forests weak when trained on a single site
- Not enough data at one location, and available features not enough
- Limited usefulness – why model a site you have a live sensor at?

Because of this, we focused entirely on our goal of modeling “virtual sensors”, i.e. nitrate concentrations/violations at locations without physical sensors.

Model	Avg CV R^2	Test R^2	Test RMSE
Dummy (mean)	-0.043051 $\pm$ 0.042876	-0.000986	1.805314
Lasso	-0.039181 $\pm$ 0.037624	-0.000986	1.805314
Ridge	-0.019393 $\pm$ 0.081711	-0.198871	1.975718
Linear Regression	-0.020238 $\pm$ 0.082605	-0.200271	1.976871
Gradient Boosting	-0.120848 $\pm$ 0.239267	-1.016990	2.562657
Random Forest	-0.020637 $\pm$ 0.167117	-1.600800	2.909995

Modeling Journey: XGBoost Model

Pooling data from all testing sites and switching to XGBoost was our first positive signal.

XGBoost with

- No tuning
- Only basic features (pre feature engineering)
- Default holdout strategies

Gave us encouraging results at unseen sites:

Regression: 0.2441 R^2

Classification: 0.51 pr-auc

These became our baselines.





Validation Strategy

Sensors on the same river report correlated values → naive splits may leak information

- In addition to Leave-One-Site-Out (LOSO) Cross-Validation, we also used Leave-One-Family-Out (LOFO) CV:
 - LOFO: sensors with overlapping basins held on same side of test/train split
 - LOSO: ignores basin overlap
- When comparing, LOFO and LOSO scores end up close — evidence cross-site leakage isn't a big practical issue — but use LOFO as the honest number



Findings: Flagging violations

The classification model is fairly decent

- **ROC-AUC** score indicates the model is good at ranking days based on their probability of violation,
- **PR-AUC** score compares the model to chance
- **Brier** is a MSE of the probabilities, so ≈ 0 means the predictions spit out by the model should be near the actual probabilities of violation

Metric	Score	Reading
ROC-AUC	0.82	The model ranks violation vs. safe days well
PR-AUC (base 0.26)	0.63	The model is 2.4 x better than chance at flagging violations
Brier	0.137	Probabilities are trustworthy



Findings: Predicting exact concentration

Predicting the exact number is hard (but we still have positive predictability)

- **LOFO $R^2 \approx 0.33$:** The model explains about a third of daily variance on unseen basins
- **Within-basin R^2 0.42:** R^2 after removing each site's mean, means model tracks day-to-day movement within a site well
- **Between-basin R^2 :** R^2 of actual vs. predicted site means. Model not good at guessing the baseline levels of sites.
- **Bottom line:** strong at "when and roughly how bad," not a calibrated meter

Metric	Score
LOFO R^2	0.33
Within Site R^2	0.42
Between-Basin R^2	0.21



Findings: The features

The model leans on physically sensible signals

- Land near the sensor matters more than distant land; rain lag matters more than same-day rain
- Clear spring/early-summer spike
- Clear important backbone features: statewide nitrate level that day, sensor location/distance geometry, near-field corn

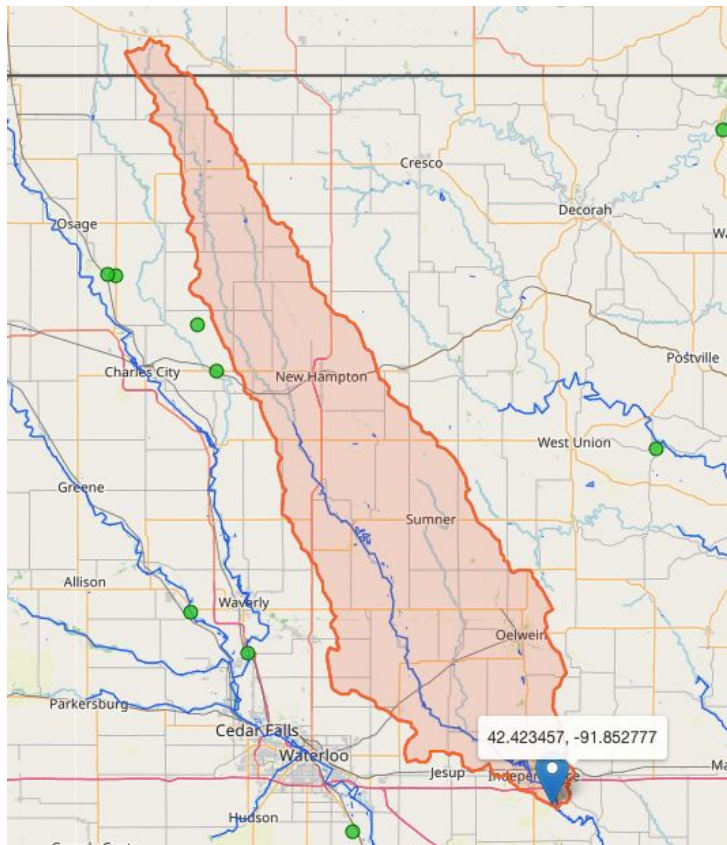


Deployment Examples

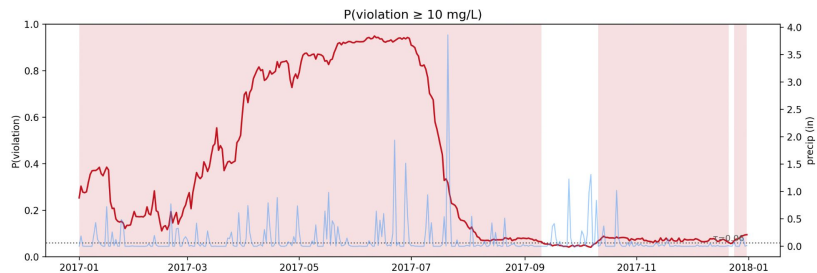
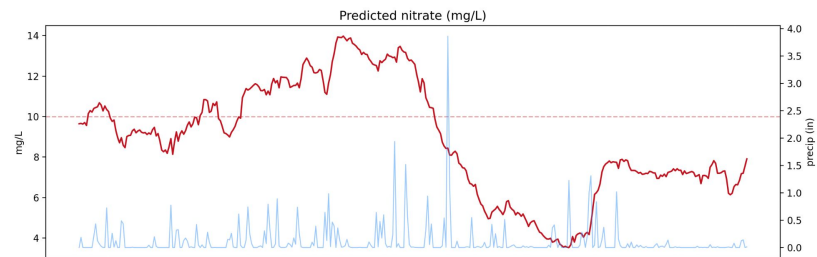
We've provided an example application in a browser-based widget.

- Drop a pin anywhere within a map of Iowa, choose a sample year
- Model computes forecasted nitrate concentration and violation days
- We provide a tuning parameter to change the balance of Actual Violations caught vs. False Alarms (True vs. False positives).

Here are examples at two sites and different years:

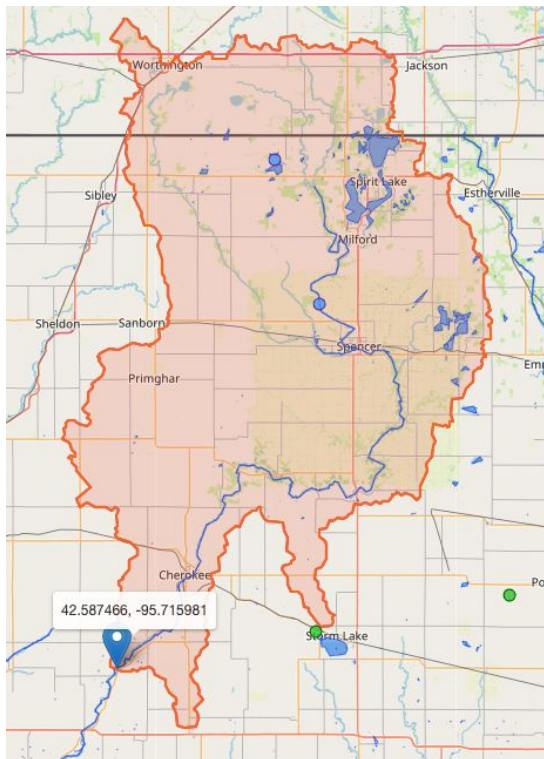


Predictions for 42.42°N, 91.85°W using 2017 example data

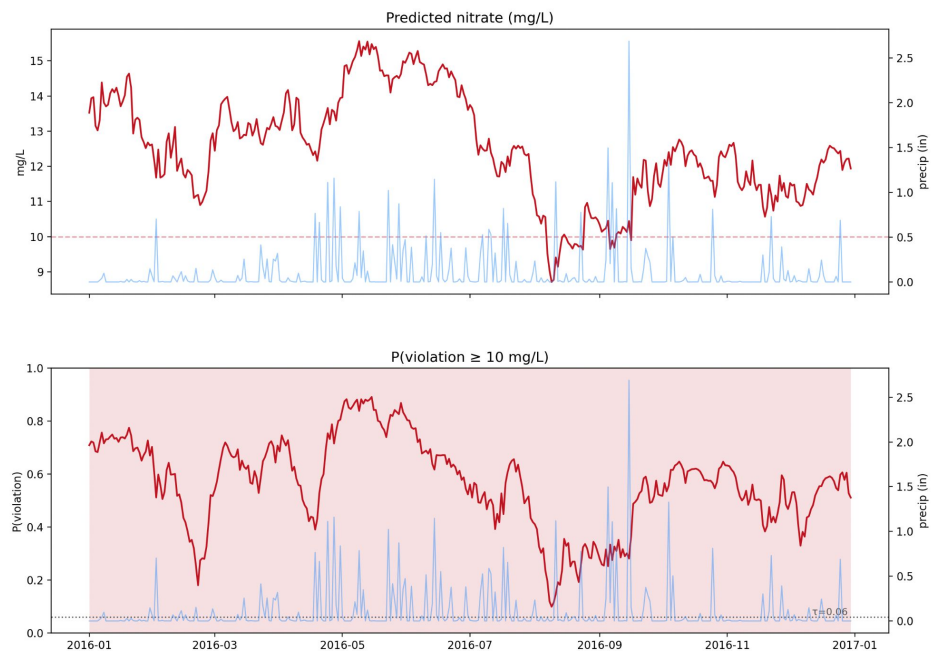


Raises alarm when violation probability ≥ 0.06 -- 127 days predicted ≥ 10 mg/L
Expected: catches ~86% of violations, ~59% of alarms false

Expect 59 % False Alarms and catch 86 % of violations
(based on training Cross-Validation)



Predictions for 42.59°N, 95.72°W using 2016 example data



Raises alarm when violation probability ≥ 0.06 -- 344 days predicted $\geq 10 \text{ mg/L}$
 Expected: catches ~86% of violations, ~59% of alarms false

Expect 59 % False Alarms and catch 86 % of violations
 (based on training Cross-Validation)



The Bigger Picture

- **Limitations**
 - sparse sensor fleet relative to Iowa's scale
 - our data is quite limited, only annual snapshots of crops and surplus
 - weather + crops + surplus isn't the whole story!



The Bigger Picture

- Next steps
 - **point source pollution data:** livestock manure a big missing link, the worst site is right next to an airport maybe that contributes as well?
 - + soil composition, tile drainage, fertilizer timing would be good additions to the dataset
 - **graph-based models** that explicitly capture river-network connectivity between basins



Business and decision relevance

Who acts on this



Water utilities

Start treatment
before a spike hits



Agencies

Target dollars at
unmonitored blind spots



Farmers

Time fertilizer
around forecasted rain

Where budget cuts leave no data, an imperfect prediction beats none